



## TP Driver Porting Guide for MTK

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|---------------------------------|-----------------------|
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## 1 Abstract

This guide introduces the structure and functions of “Hynitron TP Driver” based on linux environment and a porting reference for your MTK platform, step by step.

Through this guide, you can get how to porting “Hynitron TP Driver” to Qualcomm platform, containing Makefile/DTS modifying, kernel configuration and compiling.

You can also customize your functions of TP, containing MULTI-TOUCH protocol, GESTURE en/dis, ESDCHECK en/dis, FACTORY TEST in driver en/dis, FIRMWARE UPGRADE and so on.

Note: Examples used in the guide are verified in MTK HELLO X20 platform, only a reference, maybe different with your platform; please refer to your platform's document when porting the driver into your platform.

## 2 Interface Setting Recommendation

### 2.1 I2C interface

I2C speed recommendation: 400K (For all Hynitron's IC)

### 2.2 SPI interface

SPI speed recommendation: 8M(For CST6x IC)

## 3 File Structure

Table 1 shows the detail description of driver files

| Component | Files                                 | Attribute | Descriptor                                                                                                                                                                                                                            |
|-----------|---------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complier  | Makefile                              |           | Use for kernel compiling and configuration.                                                                                                                                                                                           |
| Main      | hyn_cfg.h<br>hyn_core.c<br>hyn_core.h |           | The main function of TP driver, that containing driver registration, bus interface (SPI/I2C) initialization, suspend/resume, multi-touch protocol support and so on. You can customize functions of TP driver by modifying hyn_cfg.h. |
| Interface | hyn_i2c.c                             |           | Bus communication with I2C/SPI;                                                                                                                                                                                                       |

|          |               |  |                                                             |
|----------|---------------|--|-------------------------------------------------------------|
|          | hyn_spi.c     |  |                                                             |
| esdcheck | hyn_ts_ext.c  |  | Use to process ESD check function.                          |
| gesture  | hyn_gesture.c |  | Use to process gesture function.                            |
| proc     | hyn_fs_node.c |  | Create sysfs/proc node, use to communicate with APK or ADB. |
| chip     | Hyn_cst***.c  |  | Use for upgrade, factory test, report point.                |

## 4 Porting TP Driver to MTK Platform

### 4.1 Copy driver files into kernel

Copy "hyn\_driver\_release" directory into kernel directory(kernel/drivers/input/touchscreen/mediatek/);  
Then modify Makefile:  
Add the line below to kernel/drivers/input/touchscreen/ mediatek /Makefile:  
obj-y += hyn\_driver\_release/

### 4.2 Configure DTS

MTK dts file path: kernel/arch/arm64/boot/dts/ amt6797\_evb\_m.dts

#### I2C Interface Example:

```
&i2c1{
    #address-cells = <1>;
    #size-cells = <0>;
    status = "okay";

    hynitron@5A {
        compatible = "hyn,66xx";
        reg = <0x5A>;
        interrupt-parent = <&tlmm>;
        interrupts = <65 0x02>;

        pos-swap = <1>;
        posx-reverse = <0>;
        posy-reverse = <0>;
    };
};
```

**SPI Interface Example:**

```
&spi0 {  
    hynitron_spi@0 {  
        compatible = "hyn,66xx";  
        reg = <0>;  
        spi-max-frequency = <6000000>;  
    };  
};
```

You can also refer to readme.txt for detail description.

## 4.3 Compile kernel and generate boot.img

```
$ make bootimage -j64
```

## 5 Driver Customization

You can modify the file of hyn\_cfg.h to customize your driver components.

**I2C\_PORT**

Select I2C or SPI interface.

**I2C\_USE\_DMA**

Set I2C transfer mode.

**HYN\_UART\_DEBUG\_EN**

Enable/Disable debug log.

**HYN\_APK\_DEBUG\_EN**

Enable/Disable debug node.

**HYN\_GESTURE\_EN**

Enable/Disable gesture function.

**MAX\_POINTS\_REPORT**

Set the maximum number of points supported.

**ESD\_CHECK\_EN**

Enable/Disable ESD check function.

**HYN\_MT\_PROTOCOL\_B\_EN**

## 5.1 Upgrade function

How to upgrade TP firmware

- a.Push firmware file(app.bin) into /sdcard/
  - > adb root
  - > adb remount
  - > adb push app.bin /sdcard/
- b.Use adb command to run upgrade
  - > adb shell
  - > echo fd>/sys/hynitron\_debug/hyntpdbg

## 5.2 Factory test function

How to do factory test

- a.Push test configuration file(hyn\_fac\_test\_cfg.ini) into /sdcard/
  - > adb root
  - > adb remount
  - > adb push hyn\_fac\_test\_cfg.ini /sdcard/
- b.Use adb command to run factory test and get test result.
  - > adb shell
  - > cat /sys/hynitron\_debug/hynselftest

## 6 Sysfs Nodes

### 6.1 Where are these sysfs nodes

a.I2C interface

When interface is I2C, then sysfs nodes will be generated under /sys/bus/i2c/devices/\*-005a/directory, "\*" stands for I2C bus number depending on your project design.

b.SPI interface

Similarly, when interface is SPI, then sysfs nodes will be generated under /sys/bus/spi/devices/spi\*/ directory, "\*" stands for SPI bus number depending on your project design.

**Warning: Sysfs nodes may be different with different driver version**

## 6.2 Component Nodes

**Gesture Component:** hyn\_gesture\_mode which are generated when set HYN\_GESTURE\_EN to 1.

node:

I2C: /sys/devices/platform/soc/\*.i2c/i2c-\*/\*/hyn\_gesture\_mode

SPI: /sys/devices/platform/soc/\*.spi/spi\_master/spi\*/spi\*/hyn\_gesture\_mode

Usage:

```
# echo 0 > hyn_gesture_mode // Disable Gesture
# echo 1 > hyn_gesture_mode // Enable Gesture
# cat hyn_gesture_mode // Show Current Gesture Status
```

## 6.3 Debugging Nodes

Sysfs debugging nodes are used to get TP driver information, and debug TP driver, which makes us debug TP driver easier and more convenient.

node: /sys/hynitron\_debug/hyntpfwver

Usage:

```
# cat hyntpfwver // show firmware version
```